

Artificial Intelligence

Instructor: Dr. Saptarshi Jana

Class XI

Assignment 1-3

Due Date: 20.06.26

30 marks

INSTRUCTIONS

- *Don't copy from AI apps*
 - *Any remaining topics will be covered during the summer break.*
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Assignment 1 Artificial Intelligence

Question A1: Who is known as the Father of Artificial Intelligence?

(1 mark)

Note: Know about contributions of all of them. Then choose the right option

- | | |
|-----------------|---------------------------|
| (a) Alan Turing | (b) John McCarthy |
| (c) Josef Capek | (d) Joseph F. Engelberger |

Question A2: What is 'Broad AI'?

(1 mark)

Note: Choose the right option.

- | | |
|----------------|----------------------|
| (a) Strong AI | (b) Weak AI |
| (c) General AI | (d) Both (a) and (c) |

Question A3: Which of these are true about AI:

- (i) AI can lead to job displacement of human workers.
- (ii) AI skills can be useful to get better jobs.

(1 mark)

Select the correct answer from below

- | | |
|-----------------------------|-----------------------------|
| (a) (i) true and (ii) false | (b) (i) false and (ii) true |
| (c) (i) and (ii) true | (d) (i) and (ii) false |

Question A5: Write five key milestone of AI.

(2 mark)

Question A6: Compare Human Intelligence and Artificial Intelligence in at least five key areas. *(3 mark)*

Question A7: Discuss the applications of AI in any three industries and explain how it transforms operations. (2 marks)

Assignment 2
Role of data and information, evolution computing

Question B1: We identify qualitative data by identifying: (1 mark)
Note: Choose the right option.

- | | |
|-------------------------|-------------------------|
| (a) Color information | (b) Emotion information |
| (c) Texture information | (d) All of these |

Question B2: Which among these is the example of structured data? (1 mark)
Note: Choose the right option.

- | | |
|------------|------------|
| (a) Tables | (b) Image |
| (c) Videos | (d) Audios |

Question B3: The full form of EDA is _____. (1 marks)
Note: Choose the right option.

- | | |
|-------------------------------|----------------------------|
| (a) Exploratory Data Analysis | (b) Explicit Data Analysis |
| (c) Explicit Design Algorithm | (d) None of these |

Question B3: How do you identify if a dataset is relevant and complete for your AI project? (2 mark)

Question B4: Describe three major types of data: structured, unstructured, and semi-structured with examples. (2 mark)

Question B5: Explain the process of data acquisition. Mention different methods and tools used. (3 mark)

Assignment 3

Data representation and programming in Python

Question C1: Python Programming: Create a list of 10 students in your class and sort them in alphabetical order. *(3 marks)*

Question C2: Write a program that takes a string as input and returns the count of vowels (a, e, i, o, u) in the string. *(3 marks)*

Question C3: Write four common types of errors in Python. *(4 marks)*